Query · src90m → 6h / 10h

One question is a vector in $\mathbb{R}^{768}$. Each time $t \in \{1, 2\}$ draws its own 128 gene questions, then appends 1 cell question.

1 Which genes

64 uniform from genes in x^{src}

48 deterministic, absent from x^{src}

16 weighted-random, absent from x^{src}

$w^{\text{top}}(g) = p(g)^{1/2} + 0.02$, arg top-48

$w^{\text{rand}}(g) = (1 - p(g))^{1/2} + 0.02$

$p(g) = \widehat{\Pr}[g \text{ on} \mid \text{type}, t]$

this cell's own target list is not used

fresh draws at $t = 1$ and $t = 2$

2 Source rank

$r_{\text{src}}(g) =$ index of g in x^{src}

$R[r] \in \mathbb{R}^{768}$, learned

table $R \in \mathbb{R}^{512 \times 768}$

rows $0, \ldots, L_s - 1$ are real ranks, $L_s \le 512$

$r = -1$ uses the last row, one shared code

unchanged from the base config

3 Target time

$T[t] \in \mathbb{R}^{768}$, learned

table $T \in \mathbb{R}^{3 \times 768}$

rows 1, 2 are 6h, 10h

90m is the source, not predicted

same vector added to every question

encoder's own time index stays fixed at 1

$$q \;=\; E[g] \;+\; R[\, r_{\text{src}}(g) \,] \;+\; T[t] \;\in\; \mathbb{R}^{768}$$

$E[g] \in \mathbb{R}^{768}$ is g's row in the encoder token table. Context stays in H_{src}, which the predictor reads as memory.

In the source

51 of the 64

- $E[g]$ — token row
- $R[r]$ — its own rank
- $T[t]$ — this time

Mask-CE

13 of the 64

- m — learned, no name
- $R[r]$ — rank stays
- $T[t]$ — time stays

Top-common absent

48, deterministic

- $E[g]$ — name stays
- $R_{\varnothing}$ — the last row
- $T[t]$ — this time

Random absent

16, weighted

- $E[g]$ — name stays
- $R_{\varnothing}$ — the last row
- $T[t]$ — this time

Cell question

slot 129

- c — learned, no gene
- – — no rank
- $T[t]$ — time only

What enters the predictor

$$[\, q_1, \ldots, q_{128},\; q_{\text{cell}} \,] \in \mathbb{R}^{B \times 129 \times 768}$$

$q_{\text{cell}} = c + T[t]$, with $c \in \mathbb{R}^{768}$. The 13 mask-CE slots are the questions that train $\mathcal{L}_{\text{mask}}$.